

Geometry

Unit 12

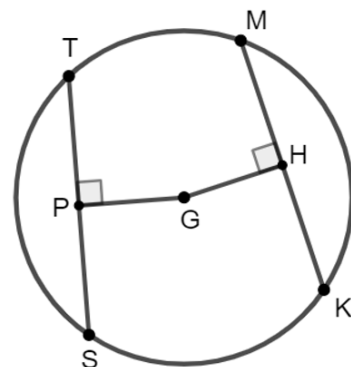
Lesson 1 Homework

Name Answer Key

Date _____

Period _____

Circle G with points T, M, K , and S on the circle are shown with chords $\overline{TS}$ and $\overline{MK}$ equidistant from the center. $TS = (12x - 5)$, $MK = (8x + 13)$, and $m\angle TPG = m\angle MHG = 90^\circ$.



1. Find the value of x .

$$12x - 5 = 8x + 13$$

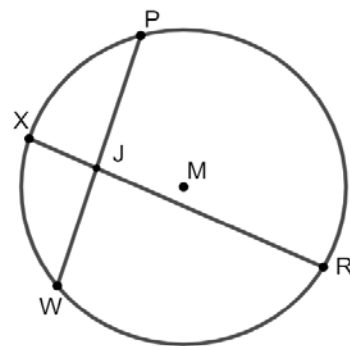
$$4x = 18$$

$$x = 4.5$$

2. What is the length of chord $\overline{TS}$?

$$12(4.5) - 5 = TS = 49$$

In Circle M , chords $\overline{RX}$ and $\overline{PW}$ intersect at point J . Points R, W, X , and P lie on Circle M . $RJ = 4y$, $JX = 3$, $PJ = (2y - 2)$, and $JW = 8$.



3. Solve for the value of y .

$$3(4y) = 8(2y - 2)$$

$$12y = 16y - 16$$

$$4y = 16$$

$$y = 4$$

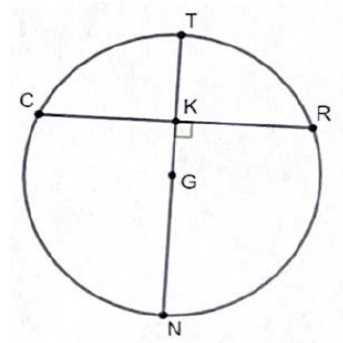
4. What is the length of $\overline{PJ}$?

$$2(4) - 2 = PJ = 6$$

5. Find the length of chord $\overline{PW}$.

$$PJ + JW = 6 + 8 = PW = 14$$

Circle G is shown where points C, T, R , and N are on the circle. The diameter, $\overline{NT}$, is a perpendicular bisector of chord $\overline{CR}$. $CR = (13x + 5)$ and $KR = (25x - 1)$.



6. Calculate the value of x .

$$2(25x - 1) = 13x + 5$$

$$50x - 2 = 13x + 5$$

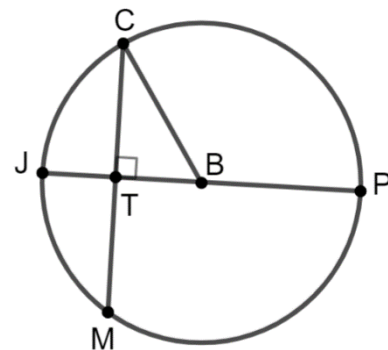
$$37x = 7$$

$$x = \frac{7}{37} \approx 0.189$$

7. What is the length of chord $\overline{CR}$?

$$13\left(\frac{7}{37}\right) + 5 = CR = \frac{276}{37} \approx 7.459$$

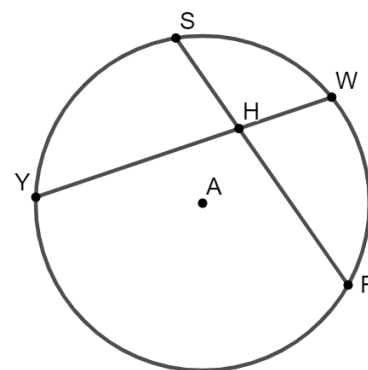
Chord $\overline{CM}$ is drawn in Circle B so that it intersects the diameter $\overline{JP}$.



8. Given that $CM = 30$ and $TB = 8$, what is the length of $\overline{BP}$?

$$BP = BC = \sqrt{(CT)^2 + (TB)^2} = \sqrt{(15)^2 + (8)^2} = 17$$

Circle A is shown where points Y, S, W , and F lie on the circle. Chords $\overline{SF}$ and $\overline{YW}$ intersect at point H . $YH = 8$, $SH = 6$, $HW = (5x - 3)$, and $HF = (2x + 10)$. Aniyah and Colton are solving for x and their work is shown below.



Aniyah's Work	Colton's Work
$8(5x - 3) = 6(2x + 10)$ $40x - 24 = 12x + 60$ $28x = 84$ $x = 3$	$6(5x - 3) = 8(2x + 10)$ $30x - 18 = 16x + 80$ $14x = 98$ $x = 7$

9. Whose work do you agree with? Justify your answer.

Aniyah is correct. Colton did not multiply the correct segment lengths.

In Circle H , points P, L, M , and S lie on the circle. Chord $\overline{PM}$ intersects the diameter $\overline{SL}$ at point G , and $m\angle HGP = 90^\circ$.

10. If $PM = 24$ and $HS = 14$, what is the length of $\overline{GH}$? Round to the nearest hundredth if necessary.

$$GH = \sqrt{(14)^2 - (12)^2} = \sqrt{52} \approx 7.21$$

